

Trainings ▾

Driveability/Low Power - Customer Complaint Form

Customer Name/Company _____ Date _____

1. How did the problem occur? Suddenly _____ Gradually _____
2. At what hour/mileage did the problem begin? Hours _____ Miles _____ Since New _____
 - After engine repair? Yes _____ No _____
 - After equipment repair? Yes _____ No _____
 - After change in equipment use? Yes _____ No _____
 - After change in selected programmable parameters? Yes _____ No _____
 - If so, what was repaired and when? _____
 - Does the vehicle also experience poor fuel economy? Yes _____ No _____

Answer questions 4 through 8 using selections (A through F) listed below. Circle the letter or letters that best describe the complaint.

- A - Compared to fleet
B - Compared to competition
C - Compared to previous engine
D - Personal expectation
E - Will **not** pull on hill
F - Will **not** pull on flat terrain
A B C D E F

- A - Can the vehicle obtain the expected road speed? Yes _____ No _____
B - What is desired speed? rpm/mpH _____
C - What is achieved speed? rpm/mpH _____
D - Gross Vehicle Weight _____
A B C D

Is the vehicle able to pull the load? Yes _____ No _____

When?

- _____ On hilly terrain
_____ With a loaded trailer
_____ On flat terrain
_____ Other _____

IF QUESTION 4 OR 5 WAS ANSWERED NO, FILL OUT THE DRIVEABILITY/LOW-POWER/EXCESSIVE FUEL CONSUMPTION CHECKLIST AND GO TO THE LOW-POWER SYMPTOM TREE.

Is the vehicle slow to accelerate or respond? Yes _____ No _____

From a stop? Yes _____ No _____

After a shift? Yes _____ No _____ rpm _____

Before a shift? Yes _____ No _____ rpm _____

No shift? Yes _____ No _____ rpm _____

Does the vehicle hesitate after periods of long deceleration or coasting? Yes _____ No _____ rpm _____

IF QUESTION 6 OR 7 WAS ANSWERED YES, FILL OUT THE DRIVEABILITY/LOW-POWER/EXCESSIVE FUEL CONSUMPTION CHECKLIST AND GO TO THE POOR ACCELERATION/RESPONSE SYMPTOM TREE.

Additional Comments:

This page can be copied for convenience.

Driveability - General Information

Driveability is a term that in general describes vehicle performance on the road. Driveability problems for an engine can be caused by several different factors. Some of the factors are engine-related and some are **not**.

Before troubleshooting, it is important to determine the exact complaint and whether the engine has a real driveability problem or if it simply does **not** meet driver expectations. The Driveability-Low-Power Customer Complaint Form is a valuable list of questions that **must** be used to assist the service technician in determining what type of driveability problem the vehicle is experiencing. Complete the checklist before troubleshooting the problem. The form can be found at the end of this section. If an engine is performing to factory specifications but does **not** meet the customer's expectations, explain to the customer that nothing is wrong with the vehicle and why.

The troubleshooting symptom charts have been set up to divide driveability problems into two different symptoms: Engine Power Output Low and Engine Acceleration or Response Poor.

Low power is a term that is used in the field to describe many different performance problems. However, in this manual low power is defined as the inability of the engine to produce the power necessary to move the vehicle at a speed that can be reasonably expected under the given conditions of load, grade, wind, and so on. Low power is usually caused by the lack of fuel flow that can be caused by any of the following factors:

- Lack of full travel of the throttle pedal
- Failed boost sensor
- Excessive fuel inlet, intake, exhaust, or drainline restriction
- Loose fuel pump suction lines.

Low power is **not** the inability of the vehicle to accelerate satisfactorily from a stop or the bottom of a grade. Refer to the Engine Power Output Low troubleshooting symptom tree in Section TS for the proper procedures to locate and correct a low-power problem. The chart starts off with basic items that can cause lower power.

Poor acceleration or response is described in this manual as the inability of the vehicle to accelerate satisfactorily from a stop or from the bottom of a grade. It can also be the lag in acceleration during an attempt to pass or overtake another vehicle at conditions less than rated speed and load.

Poor acceleration or response is difficult to troubleshoot since it can be caused by factors such as:

- Engine- or pump-related factors
- Driver technique
- Improper gear shifting
- Improper engine application
- Worn clutch or clutch linkage.

Engine-related poor acceleration or response can be caused by several different factors such as:

- Failed boost sensor
- Excessive drainline restriction
- Throttle deadband.

Refer to the Engine Acceleration or Response Poor troubleshooting symptom tree in Section TS for the proper procedures to locate and correct a poor acceleration or response complaint. For additional information, see Troubleshooting Driveability Complaints, Bulletin Number 3387245.

Engine Noise Diagnostic Procedures - General Information

Note : When diagnosing engine noise problems, make sure that noises caused by accessories, such as the air compressor and power take-off, are not mistaken for engine noises. Remove the accessory drive belts to eliminate noise caused by these units. Noise will also travel to other metal parts not related to the problem. The use of a stethoscope can help locate an engine noise.

Engine noises heard at the crankshaft speed, engine rpm, are noises related to the crankshaft, rods, pistons, and piston pins. Noises heard at the camshaft speed, one-half of the engine rpm, are related to the valve train. A handheld digital tachometer can help determine if the noise is related to components operating at the crankshaft or camshaft speed.

Engine noise can sometimes be isolated by performing a cylinder cutout test. If the volume of the noise decreases or the noise disappears, it is related to that particular engine cylinder.

There is **not** a definite rule or test that will positively determine the source of a noise complaint.

Engine-driven components and accessories, such as gear-driven fan clutches, hydraulic pumps, belt-driven alternators, air-conditioning compressors, and turbochargers, can contribute to engine noise. Use the following information as a guide to diagnosing engine noise.

Main Bearing Noise

- (Refer to the Engine Noise Excessive - Main Bearing troubleshooting symptom tree in Section TS)
- The noise caused by a loose main bearing is a loud, dull knock heard when the engine is pulling a load. If all main bearings are loose, a loud clatter will be heard. The knock is heard regularly every other revolution. The noise is the loudest when the engine is lugging or under heavy load. The knock is duller than a connecting rod noise. Low oil pressure can also accompany this condition.
- If the bearing is **not** loose enough to produce a knock by itself, the bearing can knock if the oil is too thin or if there is no oil on the bearing.

- An irregular noise can indicate worn crankshaft thrust bearings.
- An intermittent, sharp knock indicates excessive crankshaft end clearance. Repeated clutch disengagements can cause a change in the noise.

Connecting Rod Bearing Noise

- (Refer to the Engine Noise Excessive - Connecting Rod troubleshooting symptom tree in Section TS)
- Connecting rods with excessive clearance will knock at all engine speeds under both idle and load conditions. When the bearings begin to become loose, the noise can be confused with piston slap or loose piston pins. The noise increases in volume with engine speed. Low oil pressure can also accompany this condition.

Piston Noise

- (Refer to the Engine Noise Excessive - Piston troubleshooting symptom tree in Section TS)
- It is difficult to tell the difference between piston pin, connecting rod, and piston noise. A loose piston pin causes a loud double knock that is usually heard when the engine is idling. When the injector to this cylinder is cut out, a noticeable change will be heard in the sound of the knocking noise. However, on some engines the knock becomes more noticeable when the vehicle is operated on the road at a steady speed.

Oil Consumption

In addition to the information that follows, a service publication is available entitled Technical Overview of Oil Consumption, Bulletin Number 3379214.

Cummins Engine Company, Inc. defines "acceptable oil usage" as outlined in the following table.

Accetable Oil Usage									
Any Time During Coverage Period									
Engine Family	Hours per Quart	Hours per Liter	Hours per Imperial Quart	Miles per Quart	Miles per Liter	Miles per Imperial Quart	Kilometers per Quart	Kilometers per Quart	Kilometers per Imperial Quart
A	10.0	10.6	12.0	400	425	475	650	675	775
4B	10.0	10.6	12.0	400	425	475	650	675	775
6B	10.0	10.6	12.0	400	425	475	650	675	775
6C	10.0	10.6	12.0	400	425	475	650	675	775
V/VT-378	4.0	4.3	5.0	-	-	-	-	-	-
V/VT-504	4.0	4.3	5.0	250	265	310	400	425	485
V/VT-555	4.0	4.3	5.0	250	265	310	400	425	485
L Series	4.0	4.3	5.0	250	265	310	400	425	485
M Series	4.0	4.3	5.0	250	265	310	400	425	485
N Series	4.0	4.3	5.0	250	265	310	400	425	485
V/VT/VTA-903	4.0	4.3	5.0	250	265	310	400	425	485
KT/KTA-19	3.0	3.2	3.75	200	210	250	320	340	390
V/VT/VTA28	2.0	2.1	2.5	-	-	-	-	-	-
KT/KTA38	1.5	1.6	1.8	-	-	-	-	-	-
KTA50	1.1	1.2	1.3	-	-	-	-	-	-

Acceptable Oil Usage (Transit Bus, Shuttle Bus, and School Bus)									
Any Time During Coverage Period									
Engine Family	Hours per Quart	Hours per Liter	Hours per Imperial Quart	Miles per Quart	Miles per Liter	Miles per Imperial Quart	Kilometers per Quart	Kilometers per Liter	Kilometers per Imperial Quart
B	10.0	10.6	12.0	200	210	240	320	340	385
C	8.0	8.5	10.0	150	160	180	240	255	290
L, M, N	4.0	4.3	5.0	100	105	120	160	170	195



Engine Lubricating Oil Consumption Report

Owner's Name	Date of Delivery			Engine Serial Number	
	Month	Day	Year		
Address	Equipment Manufacturer			Engine Model and Horsepower	
City	State/Province	Equipment Serial Number		Fuel Pump Serial Number	
Engine Application (Describe)	Oil and Filter Change Interval			Complaint Originally Registered	
	Oil	Filters		Date	Mile/Hours/Kilometers

Lubricating Oil Added

[illegible]

Last Mileage/Hours/Kilometers _____
 Minus Start Mileage/Hours/Kilometers _____
 Equals Test Mileage/Hours/Kilometers _____
 Divided by Oil Added _____
 Equals _____
 Usage Rate _____

Customer Signature	Cummins Dealer	Cummins Distributor
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Cummins Engine Company, Inc. Form 4755



Oil Consumption Report

Customer Name:	Dist/Dir:
Engine Model:	Mi/Km/Hr:
Engine Serial Number:	CPL Number:

Oil Consumption Report	
Vehicle Make/Model:	Date:
Review of maintenance history: List any previous failures that could have had a detrimental effect on cylinder component life. Failures could include fuel, coolant, and/or foreign abrasives in the oil, second ring groove beat-out, filter plugging, etc.	
Lubrication Oil Used: Brand Viscosity Change Interval (mi/km/hr)	
Combination Oil Filter: Model Element Change Interval (mi/km/hr)	
Bypass Oil Filter: Model Element Change Interval (mi/km/hr)	
Full-Flow Oil Filter: Model Element Change Interval (mi/km/hr)	
Air Cleaner: Make and Model Change Interval	
List any external engine leaks.	
Check for any internal leaks and list them. Check turbocharger seals, valve guides, air compressor, and so forth.	
Had the fuel pump been tampered with? _____What is maximum rail pressure readings? _____ If yes, the pump must be reset to factory specifications and the customer sent out to reevaluate the oil consumption rate. The eligibility requirements must be met again, also.	
Drain and refill oil pan to check dipstick markings and note findings. WARNING: Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.	
Only after above checks are completed, leaks corrected, and proper documentation completed, disassemble engine to determine cause of the failure and repair as required.	
State reason for oil consumption.	
Signed: _____	